Introduction

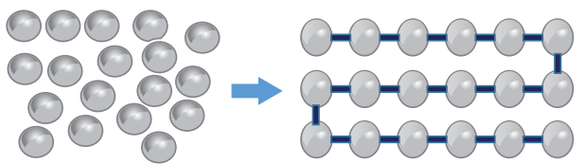
Polymer chemistry has a positive impact on your everyday life. Many of the materials you use are linked to polymer chemistry. When you get ready for school, you wear clothes, shoes, brush your teeth and take a school bag. Did you ever ask yourself what these items are made of? They are created from fibres. Fibres are made up of long chains of natural or synthetic polymers.

As a society, we are consuming more and more. The more things we buy the more things we throw away. 30 years ago, most of the waste in Tamil Nadu was biodegradable or easily recyclable. Today it is more complex. Many of the materials we use are non-biodegradable and difficult to recycle. For example, in the past people used many natural materials such as cotton, silk and jute fibres. Today we use a lot of synthetic materials such as plastics. In this lesson you will learn about polymers, different types of fibres, plastics and the waste we produce and how we should manage it.

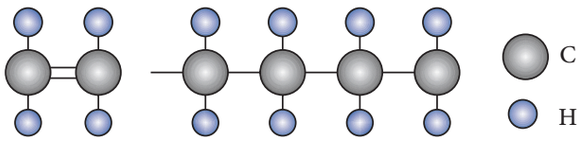
3.2 What Are Polymers?

Q4M1K1

The word 'Polymer' is of Greek origin. 'Poly' means many and 'mer' means basic smaller unit. Polymers are very long chains made of repeating smaller molecules called 'monomers' that are joined together by covalent bonds and the process is called polymerization. The diagram below shows how repeating monomers join to form a polymer:

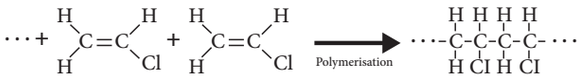


Monomers Polymer



Monomer Polymer

Polyvinyl Chloride (PVC) is a common plastic used for water pipes. The monomer and polymers of PVC is shown below.



monomer monomer polymer

ACTIVITY 1

Compare the following things. Ice cubes and a polythene bag. In both the materials, there are large number molecules combined together. Are both polymers?





3.2.1 Polymer

Polymers can be classified into natural and synthetic polymers. Can you imagine that your body produces and you are made up of natural polymers? The most familiar polymers that we use in our daily life are man-made and synthetic.

3.2.2 Natural Polymers

Natural polymers are found in living systems that include proteins and carbohydrates

VII STD Science Term-3 EM unit 3.indd 43 12-11-2019 14:43:12